GROUP 19 REPORT — RANDOM FOREST CLASSIFICATION PROJECT

1. Objective

The goal of this project is to build a supervised machine learning model to predict a binary outcome (0 or 1) from the given dataset and generate predictions for the live dataset. The focus is on achieving balanced accuracy and robustness, emphasizing correct detection of the positive class (1).

2. Data Understanding

- Dataset: Contains both numerical and categorical features with a binary target (label).
- Type: Supervised binary classification.
- Target: label $\in \{0,1\}$ — 1 represents the positive outcome.
- Live data: 6,967 rows with id and prediction.
- Checks: No duplicates; handled minor missing values; verified balanced distribution.

3. Preparation

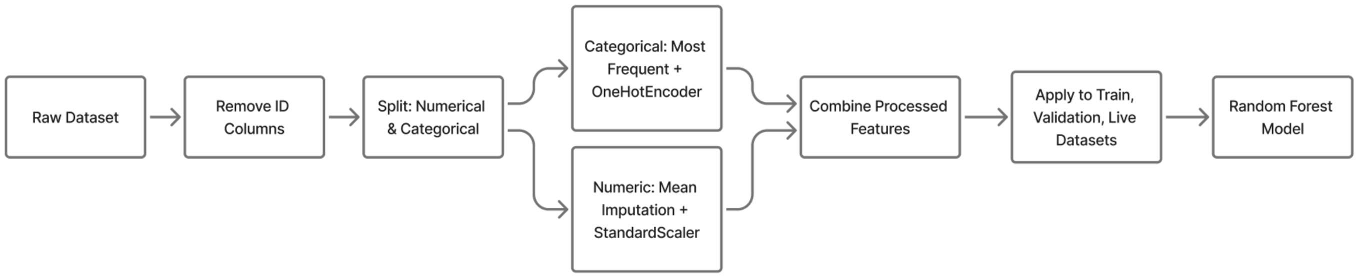
Data preprocessing used a Scikit-learn Pipeline for consistent and clean transformation.

- Removed ID columns from features.
- Numerical: Missing values → mean imputation; scaled using StandardScaler.
- Categorical: Missing values → most frequent value; encoded via OneHotEncoder.
- Applied the same preprocessing to all datasets to avoid data leakage.

4. Modeling Approach

Algorithm: Random Forest Classifier — chosen for stability and ability to model non-linear patterns.

1. Split data (80/20 train–validation).
2. Built preprocessing + model pipeline.
3. Tuned parameters using GridSearchCV: n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features.
4. Selected best model based on cross-validated F1-score, emphasizing recall for the positive class.



5. Evaluation and Results

Report performance metrics

Metric	Value
Accuracy	0.82
Precision	0.77
Recall	0.83
F1-score	0.7981

Comparison of multiple models/experiments

Model	Accuracy	F1-score	Remark
Random-Forest	0.8246	0.7981	the best performance, showing strong balance between precision and recall.
Logistic Regression	0.8119	0.7906	slightly lower, proving reliable but less capable of handling complex patterns.
Decision Tree	0.7586	0.7101	the weakest results because of overfitting and limited generalization.

6. Discussion and Analysis

- Random Forest performed best balance between accuracy and recall.
- Pipeline ensured clean, reproducible workflow.
- Feature importance can explain model behavior.
- Limitations: did not test gradient boosting; could add more hyperparameter tuning.